

Eye-Gaze Communication App

QUICK START GUIDE • 5-Minute Essentials

The patient speaks by looking LEFT and RIGHT. Each look adds one digit. Looking at the CENTER for 2 seconds finishes the word — and the tablet says it out loud.

No special hardware. No keyboard. No spelling. Just a Galaxy phone or tablet and its front camera.

The three gaze movements

Look LEFT	= 0	Adds the digit 0 to the code
Look RIGHT	= 1	Adds the digit 1 to the code
Look CENTER	2 sec	Confirms — the word is spoken aloud

Example: "Uncomfortable" = 011 → LEFT, RIGHT, RIGHT, then CENTER for 2 seconds.



LEFT (0)



RIGHT (1)



RIGHT (1)



CENTER (2 s)

Made a mistake? Just look at the center for 2 seconds. Codes that do not exist are thrown away automatically, and you can start again.

Safety: This is a communication aid, not a medical device. A caregiver must always be present, and there must always be a backup way to call for help.

1. Setting Up



Mounting position

Position the device

- About 30 cm from the patient's face
- Camera at eye level, slightly above the line of sight so it does not block the view
- Face centred in the camera preview
- Light in FRONT of the patient — never a window behind them
- Wipe the camera lens before use

Glasses: remove them if possible. If they must be worn, tilt the device until no glare shows on the lenses in the camera preview — reflections block eye detection.

First launch — 3 taps

1	Choose the language at the top of the screen: English or 한국어. This sets the menus, the word list and the voice.
2	Read the disclaimer and tap Accept.
3	On the Instructions screen choose the device type — Galaxy Phone or Galaxy Tab — so the app knows where the camera is.

The three buttons

Button	What it does
Start	Turns the camera on and starts tracking. The status light turns GREEN (TRACKING).
Pause / Resume	Suspends tracking; the camera stays on. Status light turns ORANGE (PAUSED).
Stop	Turns the camera off and stops everything.

The patient can pause without help. Closing the eyes for 5 seconds or more pauses tracking. Closing them again for 5 seconds resumes it. Use this for resting, sleeping, or while talking with visitors.

Which orientation?

- Galaxy Tab: use LANDSCAPE. In portrait the camera sits far to the side and accuracy drops.
- Galaxy Phone: either orientation works.
- On the Tab the text is about 1.7× larger and pulled toward the middle of the screen, so the eyes move less.

Before the patient starts: the caregiver should try the app first. Ten minutes of practice makes teaching much easier.

2. The Main Word Chart

This is the only page most people need. Short codes are the most urgent words — emergencies take a single look.

Code	Gaze	Says
0	L → centre	Can't breathe
1	R → centre	Suction
00	L L → centre	Yes
01	L R → centre	No
10	R L → centre	Repeat
11	R R → centre	Wait
000	L L L → centre	Thirsty ★
001	L L R → centre	Urinate ★
010	L R L → centre	Defecate ★
011	L R R → centre	Uncomfortable ★
100	R L L → centre	opens the PAIN page
101	R L R → centre	opens the ACTION page
110	R R L → centre	opens the ENVIRON · STATE page
111	R R R → centre	opens the FEELING page

Red	Emergency — one look only
Green	Yes / No / Repeat / Wait — answering questions
Yellow	★ Shortcuts — these four slots change by themselves (see below)
Blue	Doorways to the four category pages

★ The yellow shortcuts learn from use

- Every time a word is chosen from a category page, its counter goes up by one.
- The four most-used words automatically move into the yellow slots (000-011), so frequent needs get shorter codes.
- Until enough words have been used, the defaults stay: Thirsty, Urinate, Defecate, Uncomfortable.
- To see or clear the counters: Setup → Zone 3A – Usage Frequency → Show / Reset all usage counts.

Teaching order: practise the red and green words first (1–2 looks) until they are reliable. Add the yellow shortcuts next, then introduce the category pages one at a time. Sessions of 10–15 minutes work better than long ones.

3. The Four Category Pages

A category page opens after its 3-digit code. The code then restarts from empty, and a short second code picks the word. After the word is spoken the app returns to Main by itself. Looking LEFT + centre (code 0) always goes back to Main.

PAIN enter code 100 (R L L)	
0	← back to Main
1	Stomachache
00	Headache
01	Back pain
10	Neck pain
11	Chest pain
000	Nausea
001	Mouth dry
010	Eyes dry
011 - 111	empty — free to add

ACTION enter code 101 (R L R)	
0	← back to Main
1	Medicine
00	Change posture
01	Turn me
10	Lift head
11	Lower head
000	Wipe
001	Cover me
010	Call nurse
011	Come here
100	Sit
101	Stay still
110	Rest
111	Urinate

ENVIRON • STATE enter code 110 (R R L)	
0	← back to Main
1	Cold
00	Hot
01	Too bright
10	Too dark
11	Too loud
000	Sleepy
001	Itchiness
010	Thirsty
011	Uncomfortable
100 - 111	empty — free to add

FEELING enter code 111 (R R R)	
0	← back to Main
1	Thank you
00	Sorry
01	Anxious
10	Scared
11	Sad
000	Lonely
001	Happy
010	Exhausted
011	Angry
100	Miss you
101 - 111	empty — free to add

Adding your own words: Instructions screen → Setup → tap any empty *** slot → type the phrase → Save, then Save & Return. Keep phrases to 1–3 words so the voice stays clear. Three items cannot be changed: "Can't breathe" (code 0), "back to Main" (code 0 on category pages), and the four category doorways (100–111).

4. Four Worked Examples

To say	Do this
"Can't breathe"	LEFT → centre 2 s. Spoken immediately.
"Headache"	RIGHT LEFT LEFT → centre 2 s (PAIN page opens) → LEFT LEFT → centre 2 s.
"Thank you"	RIGHT RIGHT RIGHT → centre 2 s (FEELING page opens) → RIGHT → centre 2 s.
Go back to Main	LEFT → centre 2 s. Nothing is spoken.

5. If Something Is Not Working

Problem	What to do
Camera will not start	Android Settings → Apps → ALS Eye-Gaze → Permissions → allow Camera.
No voice	Check the volume buttons. Then Android Settings → General Management → Language & Input → Text-to-Speech.
Gaze not detected	More light. Clean the lens. Move the device closer. Remove glasses, or tilt the device until the glare disappears.
Reacts to every small movement	Check the eyelids are fully open, not half-closed. Ordinary small eye movements should not enter a digit.
Wrong word was spoken	Use Repeat (10) or Wait (11) to signal the caregiver, then enter the intended code again.
Unexpected words in the yellow slots	Setup → Zone 3A - Usage Frequency → Reset all usage counts.
Words in the wrong language	Go back to the Disclaimer screen and select the language again.

Safety notice

This application is a communication aid only. It is not a medical device and must never be the only way the patient can alert someone in an emergency.

- Use only with a caregiver present and monitoring.
- Always keep a backup alert method within reach.
- Have a speech-language pathologist assess the patient before long-term use.

The developers accept no liability for consequences arising from use of this application. By using it you accept these risks.

This is a short guide. The full User Manual contains the complete Setup screen reference, screen layouts for every device and orientation, and detailed training guidance.